

Multiple Choice

1. **Answer:** C

Options: A) about 18 years; B) 24 hours; C) 29 Earth days; D) a year

Note: Correct option: C - 29 Earth days

2. **Answer:** C

Options: A) The Moon will be closer to the Earth and the Earth's day will be longer.; B) The Moon will be closer to the Earth and the Earth's day will be shorter.; C) The Moon will be further from the Earth and the Earth's day will be longer.; D) The Moon will be further from the Earth and the Earth's day will be shorter.

Note: Correct option: C - The Moon will be further from the Earth and the Earth's day will be longer.

3. **Answer:** A

Options: A) 750 million years.; B) 2 billion years.; C) 3 billion years.; D) 4.5 billion years.

Note: Correct option: A - 750 million years.

4. **Answer:** D

Options: A) Titan is the only outer solar system moon with a thick atmosphere; B) Titan is the only outer solar system moon with evidence for recent geologic activity; C) Titan's atmosphere is composed mostly of hydrocarbons; D) A and D

Note: Correct option: D - A and D

5. **Answer:** C

Options: A) The first time you drop a bowling ball and it falls to the ground proving your hypothesis.; B) After you've repeated your experiment many times.; C) You can never prove your theory to be correct only "yet to be proven wrong".; D) When you and many others have tested the hypothesis.

Note: Correct option: C - You can never prove your theory to be correct only "yet to be proven wrong".

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'These craters contain the only permanently shadowed regions on Mercury'.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'black holes.'
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Solar eclipses would be much more frequent.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 300 Kelvin. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.